

Wi-Fi Roaming Experiment

1. Objective

The objective of this experiment is to understand Wi-Fi roaming by using two Access Points broadcasting the **same SSID** and observing how a wireless client moves between them.

The experiment uses:

- **Airtel Router – AP1**
- **Phone Hotspot – AP2**
- **iPhone – Roaming Client**
- **Ubuntu Laptop – Packet Capture**
- **Wireshark – Packet Analysis**

The main focus is to identify the change from one **BSSID** to another and correlate it with the client's authentication, association and WPA2 handshake.

2. Roaming Concept

A Wi-Fi network can contain multiple Access Points using the same SSID.

Each AP has its own unique **BSSID**, even though both APs advertise the same SSID.

When the client moves away from AP1 and AP2 provides a better signal, the client can move its connection towards AP2.

Therefore, the important parameters to observe in a roaming experiment are:

Same SSID + Different BSSID + Client movement + AP transition

3. Experimental Setup

The Airtel router was used as the first Access Point and the phone hotspot was configured with the **same Wi-Fi SSID and security credentials**.

The two APs had different BSSIDs:

- **AP1 – Airtel Router: FC:DD:55:B8:DA:E2**
- **AP2 – Phone Hotspot: 8E:78:4C:A6:BA:0F**

Wireless packets were captured using an Ubuntu laptop in monitor mode with Radiotap information enabled. This allowed the capture to include signal-strength information for individual frames.

4. Identifying the Two Access Points

Both APs were confirmed in the capture through their Beacon frames.

AP	BSSID	Signal
Airtel Router	FC:DD:55:B8:DA:E2	~-43.8 dBm
Phone Hotspot	8E:78:4C:A6:BA:0F	~-64.0 dBm

The different BSSIDs confirm that these are two separate wireless cells even though they use the same SSID.

```

▼ IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  ▶ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  ▶ Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Transmitter address: WeWinswirele_b8:da:e2 (fc:dd:55:b8:da:e2)
  ▶ Source address: WeWinswirele_b8:da:e2 (fc:dd:55:b8:da:e2)
  ▶ BSS Id: WeWinswirele_b8:da:e2 (fc:dd:55:b8:da:e2)
    .... .... 0000 = Fragment number: 0
    1110 0010 1001 .... = Sequence number: 3625
    Frame check sequence: 0x2dc3edd7 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
▼ IEEE 802.11 Wireless Management
  ▶ Fixed parameters (12 bytes)
  ▼ Tagged parameters (303 bytes)
    ▼ Tag: SSID parameter set: "Airtel-MyWiFi-AMF-311WW-DAE2"

```

```

▼ IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  ▶ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  ▶ Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Transmitter address: 8e:78:4c:a6:ba:0f (8e:78:4c:a6:ba:0f)
  ▶ Source address: 8e:78:4c:a6:ba:0f (8e:78:4c:a6:ba:0f)
  ▶ BSS Id: 8e:78:4c:a6:ba:0f (8e:78:4c:a6:ba:0f)
    .... .... 0000 = Fragment number: 0
    0101 1111 1010 .... = Sequence number: 1530
    Frame check sequence: 0x835fd2ac [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
▼ IEEE 802.11 Wireless Management
  ▶ Fixed parameters (12 bytes)
  ▼ Tagged parameters (250 bytes)
    ▼ Tag: SSID parameter set: "Airtel-MyWiFi-AMF-311WW-DAE2"

```

5. Client Connection to AP2

The capture first shows the client connecting to the phone hotspot.

Client MAC:

06:EC:1A:C5:3C:B1

The connection sequence is:

Authentication → Association → WPA2 Four-Way Handshake

951	14.966793722	06:ec:1a:c5:3c:b1	8e:78:4c:a6:ba:0f	802.11	90 Authentication, SN=2665, FN=0, Flags=.....C
952	14.966810615		06:ec:1a:c5:3c:b1	802.11	70 Acknowledgement, Flags=.....C
953	14.972522888	8e:78:4c:a6:ba:0f	06:ec:1a:c5:3c:b1	802.11	90 Authentication, SN=1429, FN=0, Flags=.....C
955	14.975499897	06:ec:1a:c5:3c:b1	8e:78:4c:a6:ba:0f	802.11	240 Association Request, SN=2666, FN=0, Flags=.....C, SSID="Airtel-MyWiFi- AMF-311NW-DAE2"
956	14.975525352		06:ec:1a:c5:3c:b1	802.11	70 Acknowledgement, Flags=.....C
957	14.980175330	8e:78:4c:a6:ba:0f	06:ec:1a:c5:3c:b1	802.11	202 Association Response, SN=2751, FN=0, Flags=.....C
959	14.989147206	8e:78:4c:a6:ba:0f	06:ec:1a:c5:3c:b1	EAPOL	215 Key (Message 1 of 4)
963	15.062097013	06:ec:1a:c5:3c:b1	8e:78:4c:a6:ba:0f	EAPOL	215 Key (Message 2 of 4)
964	15.062125804		06:ec:1a:c5:3c:b1	802.11	70 Acknowledgement, Flags=.....C
965	15.069443851	8e:78:4c:a6:ba:0f	06:ec:1a:c5:3c:b1	EAPOL	249 Key (Message 3 of 4)
967	15.131223340	06:ec:1a:c5:3c:b1	8e:78:4c:a6:ba:0f	EAPOL	193 Key (Message 4 of 4)

6. Client Movement Towards AP1

The client was then moved away from the hotspot and towards the Airtel router.

As the client moves, the relative signal strength of the two APs changes.

The capture contains Radiotap signal information, allowing the received signal level to be observed.

The hotspot frames are around **-64 dBm** on average, while the Airtel router is around **-43.8 dBm** in the captured environment.

```

▼ Radiotap Header v0, Length 56
  Header revision: 0
  Header pad: 0
  Header length: 56
  ▶ Present flags
  MAC timestamp: 410200337
  ▶ Flags: 0x10
  Data Rate: 54.0 Mb/s
  Channel frequency: 2437 [2.4 GHz 6]
  ▶ Channel flags: 0x00c0, Orthogonal Frequency-Division Multiplexing (OFDM), 2 GHz spectrum
  Antenna signal: -57 dBm
  ▶ RX flags: 0x0000
  ▶ timestamp information
  Antenna signal: -57 dBm
  Antenna: 0
  Antenna signal: -57 dBm
  Antenna: 1

```

```

▶ Frame 1356: Packet, 524 bytes on wire (4192 bits), 524 bytes captured (4192 b
▼ Radiotap Header v0, Length 56
  Header revision: 0
  Header pad: 0
  Header length: 56
  ▶ Present flags
  MAC timestamp: 390545290
  ▶ Flags: 0x10
  Data Rate: 1.0 Mb/s
  Channel frequency: 2437 [2.4 GHz 6]
  ▶ Channel flags: 0x00a0, Complementary Code Keying (CCK), 2 GHz spectrum
  Antenna signal: -41 dBm
  ▶ RX flags: 0x0000
  ▶ timestamp information
  Antenna signal: -50 dBm
  Antenna: 0
  Antenna signal: -41 dBm
  Antenna: 1

```

7. Client Reconnects to AP1

Later in the capture, the client connects to the Airtel router.

The Airtel BSSID is:

FC:DD:55:B8:DA:E2

The connection sequence is:

Authentication → Association → WPA2 Four-Way Handshake

```
2917 31.407118742 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... 802.11 225 Association Request, SN=2670, FN=0, Flags=.....C, SSID="Airtel-MyWiFi-AMF-311WW-DAE
2921 31.424709621 WeWinswirele_b8:da:... 6a:d5:c5:00:46:c6 802.11 297 Association Response, SN=3445, FN=0, Flags=.....C
2913 31.402169743 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... 802.11 90 Authentication, SN=2669, FN=0, Flags=.....C
2915 31.403876827 WeWinswirele_b8:da:... 6a:d5:c5:00:46:c6 802.11 90 Authentication, SN=3443, FN=0, Flags=.....C
```

```
940 14.775252841 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... 802.11 80 Deauthentication, SN=
2923 31.426719566 WeWinswirele_b8:da:... 6a:d5:c5:00:46:c6 EAPOL 215 Key (Message 1 of 4)
2925 31.459131972 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... EAPOL 215 Key (Message 2 of 4)
2927 31.462319557 WeWinswirele_b8:da:... 6a:d5:c5:00:46:c6 EAPOL 273 Key (Message 3 of 4)
2929 31.476210186 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... EAPOL 193 Key (Message 4 of 4)
```

8. Observing the AP Transition

The main evidence for the AP transition is the change in BSSID.

The two BSSIDs are:

AP1: FC:DD:55:B8:DA:E2

AP2: 8E:78:4C:A6:BA:0F

The SSID remains the same, while the BSSID changes.

9. Deauthentication and Connection Loss

The capture also contains Deauthentication frames.

For example, the hotspot shows a repeated Deauthentication sequence around frames **990–1009**.

A Deauthentication frame indicates that the current authenticated connection is being terminated.

```
1008 15.294051911 8e:78:4c:a6:ba:0f 6a:d5:c5:00:46:c6 802.11 86 Deauthentication, SN=1430, FN=0, Flags=....R...C
1009 15.294956045 8e:78:4c:a6:ba:0f 6a:d5:c5:00:46:c6 802.11 86 Deauthentication, SN=1430, FN=0, Flags=....R...C
940 14.775252841 6a:d5:c5:00:46:c6 WeWinswirele_b8:da:... 802.11 86 Deauthentication, SN=2663, FN=0, Flags=.....C
2902 31.294234223 06:ec:1a:c5:3c:b1 8e:78:4c:a6:ba:0f 802.11 86 Deauthentication, SN=2667, FN=0, Flags=.....C
```

10. Conclusion

This experiment demonstrates the basic concept of Wi-Fi roaming using two Access Points configured with the same SSID.

The Airtel router and phone hotspot were used as two different BSSs. Their BSSIDs were different, while the SSID was kept the same.

The packet capture showed the client connecting to the hotspot, experiencing different signal conditions, and subsequently establishing connectivity with the Airtel router.

The capture also contained Radiotap signal information, allowing the wireless signal level to be correlated with the client's movement.

The most important roaming indicators observed were:

Same SSID → Different BSSID → Client transition → Authentication/Association → WPA2 handshake → Data communication

This experiment helped demonstrate how the concept of Wi-Fi roaming can be investigated from actual 802.11 packet captures rather than only from the client's Wi-Fi status.